

New Kids On The Block

The new Intel Alderwood and Grantsdale chipsets offer the much awaited DDR II and PCI Express standards. Interesting features such as Graphics Media Accelerator 900, Matrix storage, High Definition Audio, Wireless Connect technology and the LGA775 (Land Grid array) CPU socket are supported by the new platforms.

The Trusted Computing Platform chip promises uncrackable DRM (Digital Rights Management).

Let's see if these new chipsets can deliver:

Chipsets

According to Intel the i915 is targeted at the masses,



The new Intel D925XCV desktop board: PCI Express will speed things up

while the i925X Express is for the performance seekers. The tables detail the various features of the new chipsets.

Memory

DDR II, running at 533 MHz, offers a total bandwidth of

Flavours of Southbridge that can be present on your boards

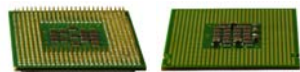
ICH6	ICH6R	ICH6W	ICH6RW
Standard	Supports onboard SATA and RAID	Supports inbuilt WiFi	Supports inbuilt SATA RAID and WiFi

8.5 GBps, which is nearly 30 per cent higher than currently available 875/865 chipset-based motherboards.

Unfortunately, this extra bandwidth is higher than the bandwidth between the CPU and the Northbridge. So, the CPU can't exploit this increased bandwidth—not until Intel's 925XE chipset, featuring support for CPUs with 1,066 MHz bus speeds.

Graphics

The age-old AGP8X bus will be replaced by PCI Express x16. Of course you'll need to upgrade your graphics card to use the new interface. The 'x16' means 16-lane memory, with each lane capable of 250 MBps speeds. This translates to speeds of 4 GBps in each direction, or a bandwidth of 8 GBps—almost four times the 2.1 GBps speed of the current 8X AGP standard. Will it ever be used? Today most graphics cards store data in huge local memories—as much as 256 MB. So,



The new CPU lacks pins

increasing bandwidth has no effect on the graphics subsystem...yet.

We ran our gaming tests



DDR II memory: life in the fast lane

on the new integrated GPU core, pitted against a Geforce4 MX card—the cheapest discrete card available. In UT2003, the Geforce4 card beat the

different purposes simultaneously—one watching a DVD movie, while the other voice conferences using headphones.

CPU

The new CPUs will be offered in LGA775 (Land Grid Array) package, which should replace the current Socket 478 gradually. The LGA775 has 775 pins as opposed to 478 on the older socket. In fact, the new CPUs don't have pins, and have flat contact points instead. You have to be careful when installing the processor, you might damage the pins on the motherboard.

Testing

Our test bed had a D915GUX motherboard, Pentium 4 550 running at 3.4 GHz, 1 GB

Onboard Graphics

Onboard graphics	Graphic Media Accelerator 900	Extreme graphics 2
Core speed (MHz)	333	266
Rendering pipelines	4	2
Pixel fill rate	1.3 gigapixel/sec	533 megapixels/sec
Pixel shader support	Pixel shader 2.0	None
Hardware T&L	None	None
Memory bandwidth available	8.5 GB/sec	6.4 GB/sec

onboard graphics by a slender margin. Other games showed similar results, proving that though the new core is better than the older Extreme Graphics 2, it's still not a true 3D solution.

Peripherals

Peripherals that earlier connected to the Southbridge, will now use the superior PCIe x1 buses.

Each device connected on this bus will have a dedicated bandwidth of 500 MBps. Now, devices such as Gigabit Ethernet, RAID controllers, etc., can overcome the bottleneck of the older bus speeds of 133 MBps.

The High Definition Audio offers 192 KHz, 24-Bit, 8-channel sound, and compliance with every possible current standard. What's interesting is the multi-streaming support; two users can use the same soundcard for two

Micron DDR2 533 MHz and two 250 GB Maxtor Maxline III hard drives. We ran Windows XP with service Pack 1 and all the latest patches.

In the Ziff Davis Content Creation test, we got 31.8, the best score we ever logged, beating the older 865 platform by nearly 12 per cent. However, in the Business Winstone test, the new platform was on par the i865-based motherboards—clearly indicating that in daily office applications, the boost is negligible.

Overall, from an Indian perspective, these platforms are still too new—they need some time to become mainstream. Though they outperform everything we've ever seen, the scarcity of DDR II memory modules, PCI Express graphics cards, power supplies, etc., will make sure that the masses have to wait before they can actually go out and buy it. ☐

	Intel 925X	Intel 915P	Intel 915G	Intel 915GV
CPUs supported	Intel Pentium 4	Intel Pentium 4 / Celeron D		
FSB frequency	800 MHz	800 / 533 MHz		
Hyper-Threading support	Supported on all			
Supported memory types	DDR2 533/400	DDR2 533/400 or DDR 400/333		
Integrated graphics	None		Intel Graphics Media Accelerator 900	
Discrete graphics slot	Yes—PCI Express X16			None
PCI Express slots	Four x1 slots			
PCI slots	6 PCI slots max.			
HDD interface	1 ATA-100 channel, 4 SATA-150 channels			
USB	Support for 8 USB 2.0 ports			
Gigabit Ethernet	Supported on all			
Audio	Intel's High Definition Audio			